

Yadagiri Avula

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Sunnyvale, California

Summary:

- **ASIC/SOC** system engineer with **13 years** of experience in Physical Design & STA.
- Experience in delivering **CPU/GPU, AI/ML, Ethernet** silicon Designs on advanced technology nodes: **2nm, 3nm, 5nm, 7nm, 10nm, 14nm, 16nm, 28nm, 40nm**.
- Experience with major foundries: **TSMC, Intel, Renesas**.
- Comprehensive knowledge on the ASIC physical design flow, from specifications to GDSII.
- Successfully delivered multiple projects from synthesis to GDSII, including several tapeouts.
- Deep expertise in timing analysis and closure at partition, section, and full-chip levels.
- Experience in Full Chip Partitioning, Bump Planning, EMIR Analysis and RDL routing.
- Supported critical projects under tight deadlines to ensure on-time tape-out.
- Skilled in updating and porting constraints as per design changes.
- Developed scripts for bulk ECOs to fix timing and design rule violations at various levels.
- Experienced in both flop-based and latch-based design implementation and timing.
- Managed multi-voltage and low-power design implementations.
- Proficient in creating, analyzing, and updating UPF files for complex SoC designs.
- Implemented H-tree and clock binary tree structures to achieve balanced clock distribution and minimize skew across large-scale integrated circuits.
- Developed and executed custom routing strategies tailored to specific design requirements, enhancing performance and reliability.
- Executed comprehensive benchmarking methodologies to assess cutting-edge semiconductor process technologies.
- Hands-on with complex, high-frequency, and macro-dominated designs, including timing and signoff closure.
- Provided technical leadership, guidance, and daily support to team members, including technical reviews and decision-making.
- Skilled at debugging and adapting workflows to accommodate design changes.
- Developing scripts to streamline repetitive processes and review design summaries.

EDA Tools:

- **Synopsys:** Fusion Compiler, ICC2, PrimeTime, ICC, Star RC, ICV.
- **Cadence:** Innovus, Tempus, Genus, Conformal LEC, SOC-FE.
- **Mentor Graphics-** Calibre, Calibredrv.
- **Ansys:** RedHawk, **Ausdia:** Time Vision.

Professional experience:

- **Domain Architect:** Wipro Limited, Mountain View, California | April 2024 - Present.
- **Technical Lead:** Wipro Limited, Hyderabad | July 2017 – March 2024.
- **IC Design Engineer II:** Broadcom Limited. Bangalore | April 2015 - July 2017.
- **Design Engineer:** Smartplay Technologies Pvt. Ltd. Bangalore | July 2014 - April 2015.
- **Associate Member of Technical Staff:** DXCorr, Bangalore | April 2013 - July 2014.
- **Assistant Professor** in Electronics Engineering | June 2009 - June 2010

Academic Qualifications:

- **M. Tech (VLSI System Design),** JNTU, Hyderabad
- **B. Tech (Electronics and Comm.),** JNTU, Hyderabad

Programming skills:

- Expertise in writing scripts using TCL, PERL, C-Shell, and Python.

Professional Affiliations:

- **IEEE - Senior Member** (Elevated 2026)

Project Details:

META, Sunnyvale California | April 2024 - present

Project: TSMC 2nm project.

Roles: Contingent Worker Supporting SOC activities and hierarchical custom block implementation.

Key Activities:

- Evaluating and implementing architectural and design changes, including feasibility studies.
- Exploring different bump structures, placement and routing challenges.
- Implementing custom hierarchical blocks from Synthesis to Routing along with timing analysis.
- Collaborating with RTL, DFT team to resolve roadblocks and finding work arounds.
- Evaluating power numbers for different bump structures.
- Enabling SOC and sub-block implementation with different bboxes and delivering collaterals.
- SOC activities such as bound analysis, repeater planning and bump planning.
- Automation for quick data Analysis and improved efficiency.

NVIDIA, Bengaluru | July 2023 – March 2024

Project: TSMC 3nm GB20B project.

Roles: Contingent Worker Responsible for SM block PnR and Timing closure.

Key Activities:

- Optimized a 2.5GHz SM chiplet, containing four sub-blocks and instantiated 50 times at chip level.
- Led multiple PnR (Place and Route) trials to enhance power, performance, and area (PPA) of SM.
- Collaborated with full-chip timing owners to resolve timing violations within SM chiplets.
- Applied targeted constraints to critical scenarios for improved PnR optimization.
- Conducted comparative PnR runs using both Synopsys and Cadence tools to identify the best PPA.
- Coordinated with methodology teams to address tool and flow issues, enabling new PnR runs.
- Assessed metal layer costs by evaluating 13 vs. 15 metal layer stacks.
- Diagnosed and resolved congestion and timing issues at the block level.
- Adjusted block size (bbox) and frequency to study partition convergence and optimize PPA.

Silicon Labs, Hyderabad | Jan 2023-June 2023

Project: Caracal 40ULP TSMC project.

Roles: Contingent Worker Full Chip timing closure across all modes.

Key Activities:

- Collaborated closely with the PnR team to resolve timing issues, analyse timing paths, and recommend floorplan changes to meet timing requirements.
- Ported constraints from 128-bit to 256-bit widths by coordinating with the design team to accommodate logic additions. Generating ECOs for Setup, Hold and DRVs.
- Successfully closed challenging "last bucket" timing violations across functional, scan, and cross-corner modes.

INTEL, Bengaluru | July 2017 – Jan 2023

Project: DG2 5nm MTL128 & MTL64, 7nm ADLN & RPLS, 7nm RWC cluster, 16nm TGM.

Roles: Contingent Worker Leading the team and owning full chip sub sections, it's partitions.

Key Activities:

- Responsible for 5 sub-sections and its partitions with 1M+ instance count.
- Co-ordinating with section and partition owners, resolving the roadblocks, showstoppers if any.
- Making sure that all sections and partitions run going fine with updated analysis and fixes.
- Working with RTL, DV, DFT, RV and LV cross functional teams to find out issues and provide fixes.
- For a few partitions PnR tasks are implemented in innovus to evaluate convergence between the tools.
- Implemented latch-based designs with 4.5GHz critical in timing and placement.

BROADCOM, Bengaluru | April 2015 - July 2017

Project: Designing EBIST memory blocks with PnR flow in 7nm, 10nm and 16nm.

Roles: Implementing the blocks with EBIST logic using IC compile.

Key activities:

- Floorplan, power plan, placement, CTS and route design to meet the timing with less congestion.
- Automate the flow for different memory compilers (single port, dual port).
- Meet the Area, power and timing requirements with a handful of DRCs.

Awards Received:

- Extraordinaire, 3 Cheers, League of Superheroes, Team Impact